

Akka vs. n8n

A comparison for teams building agentic AI

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n8n automates your workflows; Akka runs your production agentic systems — and guarantees them. n8n is excellent visual workflow automation with AI Agent nodes — the fast path for integrations and internal automations. It is not built for production agentic scale, durable execution, or AI governance: the heavy-load, reliability, and compliance burden stays with you. Akka owns all of it under a contractual SLA, and Akka Specify can deliver and run the entire governed system for you as a guaranteed outcome.

99.9999%

AKKA AVAILABILITY SLA

Weeks

AKKA SPECIFY DELIVERY

Fixed

ONE ALL-IN PRICE

90%

LESS INFRASTRUCTURE

DIMENSION	N8N	AKKA
What it is	A visual workflow-automation platform with AI Agent nodes	A full-stack agentic systems platform — build on it yourself, or have it delivered and operated for you
How you reach production	Self-build around the automation canvas, or contract a labor-led integration engagement	Build with spec-driven development, or Akka Specify delivers and runs it
Commercial model	Per-execution metering, plus build-and-operate labor for everything beyond automation	One fixed price — platform, infrastructure, tokens, training, delivery, and operations
Outcome guarantee	Effort only; the outcome is not guaranteed	The delivered outcome is guaranteed
Scope	Workflow automation and integrations; agentic scale, durable execution, and governance are the customer's burden	Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime
Availability SLA	99.5% uptime (Enterprise Cloud), excluding planned maintenance	99.9999% — entire platform, backed by indemnities
HA / DR	Queue mode + multi-main, Redis required; no published RTO/RPO or active-active multi-region guarantee	Sub-1-minute RTO; zero-byte RPO; active-active across regions
Durable execution	Execution state persisted to PostgreSQL; in-flight runs on a crashed worker can be marked "crashed" — no deterministic replay substrate	Event-sourced; the runtime handles recovery and replay
AI agents	AI Agent node built on the LangChain JS framework	Native agents with tools, handoffs, guardrails, interaction logging
Memory	Bolt-on sub-nodes; external store you provision	Durable in-memory, 4ms reads / sub-10ms writes
Governance / EU AI Act	Audit logs and log streaming (Enterprise); no inline policy enforcement, classification, or sealed audit artifact	Aspect-woven runtime enforcement + full pre-production governance
Infrastructure footprint	Per-execution metering (€24–€800/mo by plan; Enterprise custom) + infra you operate when self-hosted	Shared compute; up to 90% lower infrastructure for the same workload
Improves after go-live	Not provided	Continuous evaluation, reinforcement learning, and distillation — up to 80% lower

		token cost over time
Certifications	SOC 2 Type II, GDPR (trust.n8n.io)	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)
License	Sustainable Use License (fair-code, source-available, not open source); internal-business-purpose only; reselling n8n as a service prohibited; n8n Embed needs a commercial license	BSL; deploy anywhere including sovereign cloud; portable specs

1. n8n Is an Automation Tool; Akka Is an Agentic Platform

n8n is a visual workflow-automation platform that connects 400+ apps and APIs, with AI Agent nodes added on top. It is excellent for integrations and internal automations, and self-hosting gives teams full control of their data. A production agentic system needs more than node-based automation — durable execution, horizontal agentic scale, durable memory, streaming, and runtime governance — and n8n leaves those to you.

Capability	n8n	Akka
Visual workflow automation + integrations	Yes (400+ integrations)	Programmatic; integrations via code
Native AI agents (tools, handoffs, streaming)	AI Agent node on LangChain JS	Built in
Durable execution / replay	PostgreSQL-persisted; no deterministic replay	Event-sourced, runtime-managed recovery
Durable memory	Bolt-on sub-nodes; external store	Built in, 4ms / sub-10ms
Real-time streaming	None native	Built in, backpressured, petabyte-scale
Inline governance / policy enforcement	None	Inline, runtime-embedded
Pre-production governance	None	Classification, sign-offs, sealed posture

n8n’s AI Agent node is a wrapper over the LangChain JavaScript framework. It is ideal for triggering an LLM step inside an automation; it is not an agent runtime engineered for unattended, process-attached execution at scale.

2. Two Ways to Production: Build It, or Have It Delivered

Akka offers two ways onto one platform: build the system yourself with spec-driven development, or have **Akka Specify** deliver and operate the entire governed system for you as a guaranteed outcome — in weeks, for one fixed price. n8n offers neither a platform nor a delivery model for a production agentic system: past the automation canvas, durable execution, agentic scale, memory, streaming, and governance are the customer’s to build, run, and own.

	With n8n	With Akka Specify
Model	Self-build, or a labor-led integration engagement	You provide the specifications
Who does the work	Your engineers, or forward-deployed / staff-augmentation contractors	Akka generates, governs, delivers, and runs the system
Timeline	Months, by construction	Weeks, not quarters
Billing	Time-and-materials, plus metered platform parts	One fixed price
Afterward	You inherit the system and its operations	Kept up, safe, and improving — operated by Akka
Guarantee	Effort is guaranteed; the outcome is not	The delivered outcome is guaranteed

n8n is scoped to workflow automation: durable execution, horizontal agentic scale, memory, streaming, and governance are the customer’s responsibility to build around it, on n8n Cloud or self-hosted. A team that cannot build those layers itself contracts a labor-led integration engagement — billed for time and materials, guaranteeing effort rather than the outcome, and handing back a system the team then owns and operates. Akka Specify inverts that: you provide a handful of plain-language specifications, and Akka generates, tests, governs, deploys, and runs the system — then keeps it available, safe, and improving under one agreement.

3. Reliability, Scale, and Durable Execution

n8n Enterprise Cloud publishes a [99.5% uptime](#) commitment (excluding planned maintenance during 10am–5pm German time) — roughly 1.8 days of allowed downtime per year. Akka’s contractual SLA is 99.9999% — about 31 seconds per year — across the entire platform, backed by indemnities.

Metric	n8n	Akka
Availability SLA	99.5% (Enterprise Cloud)	99.9999%
Allowed downtime / year	~1.8 days	~31 seconds
RTO	Not published	Sub-1 minute
RPO	Not published	Zero byte
HA model	Queue mode + multi-main; Redis + PostgreSQL required	Active-active across regions
Durable execution	PostgreSQL-persisted; crashed-worker runs can be marked "crashed"	Event-sourced, replayable from the journal
Who operates it	The customer (or an inherited integrator)	Akka SREs, 24/7

To scale horizontally, n8n runs in **queue mode**: a Redis message broker (Bull queue) distributes executions to worker processes, PostgreSQL is mandatory, and multiple "main" instances provide availability. This is capable for automation throughput, but n8n provides no deterministic durable-execution substrate — when a worker dies mid-execution, in-flight runs can be marked "crashed" rather than transparently replayed, and recovery is the operator's responsibility. Akka's actor-based runtime runs 4KB actors at 200M actors/core across clusters up to 1M nodes; state is durable in-memory and replayable from the event journal; the runtime owns recovery, so application code does not have to — and Akka operates that runtime under the SLA, 24/7.

4. Up to 90% Cheaper to Operate — and It Keeps Getting Cheaper

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume, not list price. The drivers are actor concurrency (~10T tokens/core/year vs ~2T; ~80% less compute than Python frameworks; Manulife: up to 300% more concurrency, 30–50% faster processing after porting from Python), shared compute, and micro-checkpointing.

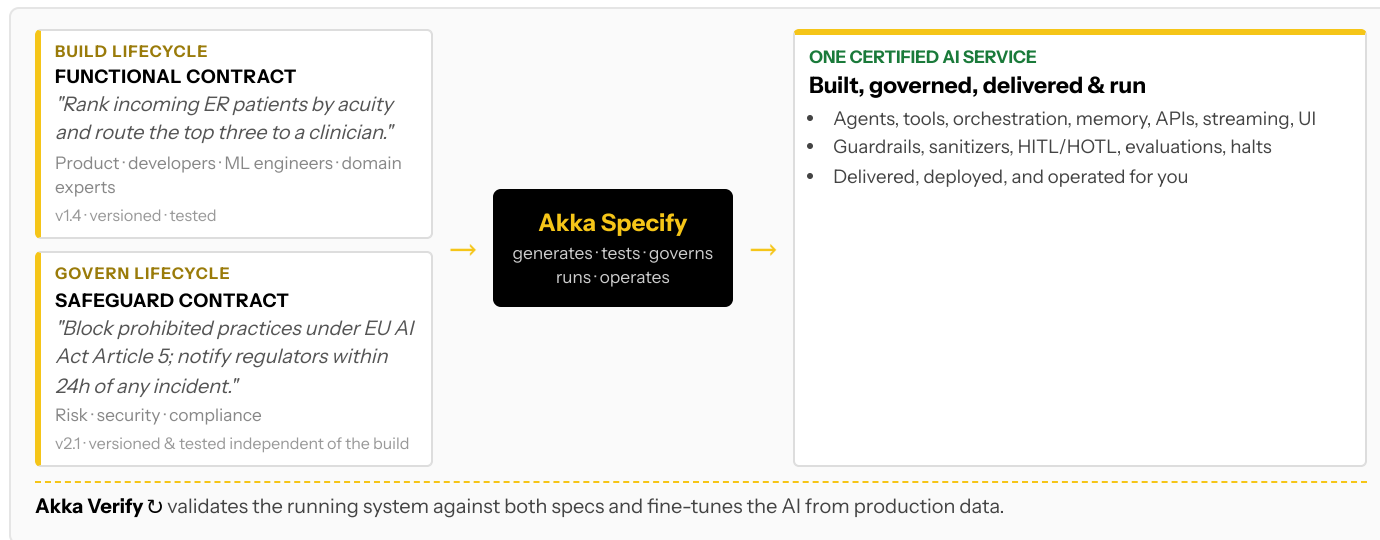
n8n Cloud bills **per execution** (€24/mo Starter / 2,500 executions, up to €800/mo Business / 40,000 executions; Enterprise custom), and that meter covers automation only. Memory, streaming, agent infrastructure, and the database sit outside it — provisioned and operated by you when self-hosted. Akka runs all of it on one shared-compute runtime for **one fixed price** — covering platform, infrastructure, tokens, training, delivery, and operations. The delivered outcome also compounds: Akka Verify's continuous evaluation, reinforcement learning, and distillation cut token cost up to 80% over time — a capability n8n does not provide.

6. How Akka Governs Inline

The EU AI Act is enforceable now (prohibited practices since Feb 2025, high-risk since Aug 2025). Akka governs at the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; pre-deployment classification against **186 regulations and 877 controls** (288 of which carry a financial penalty); multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Governance n8n leaves to you, Akka enforces inline — and delivers it already wired when the system is built through Akka Specify.

7. One Certified System — Built, Governed, Delivered, and Run

Building on n8n means a workflow author wiring nodes on a canvas; there is no co-equal, independent governance lifecycle and no path for a risk officer to author and version enforceable safeguards. Akka runs two independent lifecycles on one platform via **Akka Specify**:



Akka generates, tests, governs, **runs, and operates** one certified service from both specs, which are versioned and tested independently by different audiences. Through Akka Specify, that certified system is delivered and operated as a guaranteed outcome — a delivery model and an independent governance lifecycle n8n has no equivalent for.

8. Real-Time Streaming at Petabyte Scale

5. Governance and the EU AI Act

n8n Enterprise provides audit logs and log streaming — useful operational telemetry. It provides no AI governance for production agents: no inline policy enforcement, no decision explainability, no human pause/override of a running agent, no immutable interaction ledger, no pre-deployment classification, and no sealed audit artifact. The compliance burden stays with the customer.

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover

High-risk AI carries a 10-year logging-retention obligation (Art. 72).

n8n has no native streaming engine; real-time pipelines and agent feedback loops are provisioned separately. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

9. For the Buyer: Risk, Compliance, Accountability, and Licensing

Buyer concern	n8n	Akka
Certifications & audits	SOC 2 Type II, GDPR (trust.n8n.io)	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Delivery & outcome	Self-build or a time-and-materials integrator; effort is guaranteed, the outcome is not	Akka Specify delivers and operates the system for one fixed price; the outcome is guaranteed
Scope of accountability	The automation layer; you operate scale, durability, and governance	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Standard cloud terms	Availability and data-integrity guarantees backed by contractual indemnities
Track record & funding model	Venture-funded: \$180M Series C at a \$2.5B valuation (Oct 2025); founded 2019; scaling on investor capital	Profitable and self-funding; 18 years and 100,000+ deployments (52 banks); Dell Technologies Capital is largest shareholder, a customer, and an AI partner
Budget predictability	Per-execution metering that scales with load	One fixed price finance can forecast
License	Sustainable Use License — fair-code, source-available, not open source; internal-business-purpose only; reselling/hosting n8n as a service prohibited; embedding requires the n8n Embed commercial license	BSL; deploy on Akka cloud, hyperscaler VPC, your Kubernetes, on-prem, or sovereign cloud; portable specs

n8n is well-funded and growing fast — the decision is scope and accountability. n8n gives you a best-in-class automation canvas to integrate apps and trigger AI steps; Akka gives you the agentic platform, and owns the SLA, the durability, and the governance — and, through Akka Specify, will deliver and run the whole system for you. The Sustainable Use License is also a procurement gate worth reading early: source-available but commercially restricted, with no right to offer n8n as a hosted service and a separate commercial license required to embed it.

Customers Running Agentic and Real-Time Systems on Akka

Manulife 2,000 developers across 100 projects on one governed platform	Tubi 5B tok/s real-time hyper-personalization engine	Swiggy 71ms order-assignment AI, ~50% latency reduction	John Deere 1,000+ tractor sensors turned into real-time insight	Verizon 750% order-processing capacity gain; 6s → 2.4s response
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Common Questions

We don't have a team to build this — what are our options?

Two. Build it on the platform with spec-driven development, or have Akka Specify deliver and operate it for you. You provide plain-language specifications; Akka generates, governs, delivers, and runs the system as a guaranteed outcome, for one fixed price. With n8n, production agentic scale is self-build around the automation canvas, or a labor-led integration engagement you then own.

How is Akka Specify different from hiring an integrator to build our n8n workflows?

An integration engagement sells effort — time-and-materials over months — and hands back a system you own and operate; the outcome is not guaranteed. Akka Specify sells the outcome: a governed system delivered in weeks for one fixed price, then kept up, safe, and improving by Akka. You own the specifications, not the operational burden.

n8n has AI Agent nodes and raised \$180M at a \$2.5B valuation. Doesn't that close the gap?

n8n's AI Agent node is a LangChain-based step inside a workflow, and strong funding reflects real demand for automation. But agent nodes on an automation engine are not a production agent runtime: there is no durable-execution substrate, no inline governance, and no platform-wide SLA. Capitalization does not add those layers.

Can we add governance on top of n8n?

You can add log streaming and audit logs, but the EU AI Act expects enforcement inline to the runtime: immutable records witnessed as they happen, human override on running processes, classification before a system ships, and a sealed audit artifact. Telemetry read after the fact cannot gate a deployment. Akka embeds all of this and covers pre-deployment governance.

Isn't n8n cheaper because it is free to self-host?

We already run n8n for our automations. Why add Akka?

Keep n8n for what it is excellent at — integrations and internal automations. When you move into production agentic AI you also need durable execution, horizontal agentic scale, durable memory, streaming, and runtime governance, which n8n does not provide. Akka runs the agentic systems on one platform while n8n keeps handling your automations.

Sources

n8n Enterprise Cloud SLA: n8n.io/legal/cloud-enterprise-terms — 99.5% uptime, excluding planned maintenance (10am–5pm German time)

n8n scaling / queue mode: docs.n8n.io/hosting/scaling/queue-mode — Redis (Bull queue) + worker processes, PostgreSQL required, multi-main for availability

n8n durable execution: community.n8n.io ("What happens if n8n worker crashes?") & [github.com/n8n-io/n8n #22541](https://github.com/n8n-io/n8n/issues/22541) — crashed-worker executions can be marked "crashed"; recovery operator-managed

n8n AI Agent node: docs.n8n.io/integrations/builtin/cluster-nodes/root-nodes/n8n-nodes-langchain.agent — built on the LangChain JS framework

n8n pricing: n8n.io/pricing — per-execution metering, €24/mo Starter (2,500 exec) to €800/mo Business (40,000 exec); Enterprise custom

n8n license: docs.n8n.io/sustainable-use-license & blog.n8n.io/announcing-new-sustainable-use-license — fair-code, source-available, not open source; internal use only; no reselling as a service; n8n Embed commercial license

n8n security & certifications: n8n.io/legal/security & trust.n8n.io — SOC 2 Type II, GDPR

The n8n engine is source-available under the Sustainable Use License (fair-code, not open source), and self-hosting means you build and operate scale, HA/DR, durability, and everything around it. Cloud bills per execution. Akka's shared-compute model is up to 90% cheaper to operate than the equivalent assembled stack, on one fixed price — and the license is a procurement gate: reselling n8n as a service is prohibited and embedding requires a commercial license.

n8n funding: blog.n8n.io/series-c — \$180M Series C at \$2.5B valuation (Oct 2025); founded 2019

Akka Specify (spec-driven delivery): akka.io / akka.io/llms.txt — you provide the specifications; Akka generates, tests, governs, delivers, and operates the system as a guaranteed outcome; one fixed price covering platform, infrastructure, tokens, training, delivery, and operations; delivered in weeks; continuous improvement via reinforcement learning and distillation (up to 80% lower token cost, higher accuracy)

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Akka platform: 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); 186 regulations / 877 controls / 288 with a financial penalty; 100,000+ deployments / 18 years; profitable; Dell Technologies Capital